

Appendix: Research Poster

Simulating Voices: Does Instruction Tuning Make a Simulated Crowd Hyper-Accurate?

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1 References

Complete list of the sources consulted for the poster. The poster cites them by author name. Literature numbers on the poster are taken from these publications (page and table references are given on the poster); experiment numbers come from the author’s own experiment.

References

- [1] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901, 2020. URL <https://arxiv.org/abs/2005.14165>.
- [2] OpenAI. GPT-4 technical report. Technical report, OpenAI, 2023. URL <https://arxiv.org/abs/2303.08774>. arXiv:2303.08774.
- [3] Gati V. Aher, Rosa I. Arriaga, and Adam Tauman Kalai. Using large language models to simulate multiple humans and replicate human subject studies. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pages 337–371, 2023. URL <https://proceedings.mlr.press/v202/aher23a.html>.
- [4] Claire Wardle and Hossein Derakhshan. Information disorder: Toward an interdisciplinary framework for research and policymaking. Technical report, Council of Europe, Strasbourg, 2017. URL <https://edoc.coe.int/en/media/7495-information-disorder-toward-a-n-interdisciplinary-framework-for-research-and-policy-making.html>.
- [5] Ivan Vykopal, Matúš Pikuliak, Ivan Srba, Robert Moro, Dominik Macko, and Maria Bielikova. Disinformation capabilities of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14830–14847, 2024. doi: 10.18653/v1/2024.acl-long.793. URL <https://doi.org/10.18653/v1/2024.acl-long.793>.
- [6] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models

- to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744, 2022. URL <https://arxiv.org/abs/2203.02155>.
- [7] Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. Whose opinions do language models reflect? In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pages 29971–30004, 2023. URL <https://proceedings.mlr.press/v202/santurkar23a.html>.
 - [8] Angelo Gaeta, Vincenzo Loia, Angelo Lorusso, Francesco Orciuoli, and Antonella Pascuzzo. Towards a LLM-based intelligent system for detecting propaganda within textual content. *Computers and Electrical Engineering*, 128:110765, 2025. doi: 10.1016/j.compeleceng.2025.110765. URL <https://doi.org/10.1016/j.compeleceng.2025.110765>.
 - [9] Samuel C. Woolley and Philip N. Howard, editors. *Computational Propaganda: Political Parties, Politicians, and Political Manipulation on Social Media*. Oxford Studies in Digital Politics. Oxford University Press, 2018. doi: 10.1093/oso/9780190931407.001.0001. URL <https://doi.org/10.1093/oso/9780190931407.001.0001>.
 - [10] Ben Buchanan, Andrew Lohn, Micah Musser, and Katerina Sedova. Truth, lies, and automation: How language models could change disinformation. Technical report, Center for Security and Emerging Technology (CSET), May 2021. URL <https://doi.org/10.51593/2021CA003>.
 - [11] Rowan Zellers, Ari Holtzman, Hannah Rashkin, Yonatan Bisk, Ali Farhadi, Franziska Roesner, and Yejin Choi. Defending against neural fake news. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://proceedings.neurips.cc/paper_files/paper/2019/hash/3e9f0fc9b2f89e043bc6233994dfcf76-Abstract.html.
 - [12] Qwen Team, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, et al. Qwen2.5 technical report, 2024. URL <https://arxiv.org/abs/2412.15115>.
 - [13] Stanley Milgram. Behavioral study of obedience. *The Journal of Abnormal and Social Psychology*, 67(4):371–378, 1963. doi: 10.1037/h0040525. URL <https://doi.org/10.1037/h0040525>.
 - [14] James Surowiecki. *The Wisdom of Crowds: Why the Many Are Smarter Than the Few and How Collective Wisdom Shapes Business, Economies, Societies and Nations*. Doubleday, New York, 2004. ISBN 9780385503860. URL <https://openlibrary.org/isbn/9780385503860>.
 - [15] Dominik Macko, Robert Moro, Adaku Uchendu, Jason Lucas, Michiharu Yamashita, Matúš Pikuliak, Ivan Srba, Thai Le, Dongwon Lee, Jakub Simko, and Maria Bielikova. MULTITuDE: Large-scale multilingual machine-generated text detection benchmark. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9960–9987, 2023. doi: 10.18653/v1/2023.emnlp-main.616. URL <https://doi.org/10.18653/v1/2023.emnlp-main.616>.
 - [16] Giovanni Da San Martino, Alberto Barrón-Cedeño, Henning Wachsmuth, Rostislav Petrov, and Preslav Nakov. SemEval-2020 task 11: Detection of propaganda techniques in news articles. In *Proceedings of the Fourteenth Workshop on Semantic Evaluation*, pages 1377–1414, 2020. doi: 10.18653/v1/2020.semeval-1.186. URL <https://doi.org/10.18653/v1/2020.semeval-1.186>.

2 Use of Generative AI Tools

The following generative AI tools were used while preparing the poster:

- **Antigravity** (Google), an AI coding and writing assistant;
- **Claude Code** (Anthropic), with the model Claude Opus 5.5.

They were used to help draft and rephrase text, to check references and the literature numbers against the original publications, to write and debug the experiment code (Python), and to produce the LaTeX layout, the diagram and the plot. The experiment data are outputs of the open Qwen2.5 models [12] produced by that code; all experiment numbers on the poster are computed by the analysis script from the saved raw outputs (code and outputs: <https://github.com/ssowrabha249-coder/Simulating-Voices-and-Fabricating-Realities--Poster>). The author reviewed and edited all content and takes full responsibility for it.

Declaration of Academic Integrity

1,

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hereby declare that I have written this thesis independently and have not used any sources or aids other than those specified. All direct or indirect content and ideas from external sources are acknowledged as references. I confirm that I have not previously submitted this thesis in part or whole for any other academic credit nor has the work been submitted to any other examination office and has not been published.

I confirm that I have only used text or other content-generating aids based on generative artificial intelligence (GenAI) in the manner permitted in writing by the examiner and that I have explicitly marked GenAI-generated passages.

When using GenAI tools, I have indicated these with their product name and listed them in full in an appendix.

I am fully responsible for any GenAI-generated passages and content I have used in my work.

Thesis title: Simulating Voices: Does Instruction Tuning Make a Simulated Crowd Hyper-Accurate?

Programme: M.Sc. Natural Language Processing

Examiner: Dr. phil. Simon Werner

This is an achievement for a

- ☐ Bachelor's thesis
- ☐ Master's thesis
- ☒ other examination achievement for module (term paper etc.) Research poster, NLP module

Trier, 30.09.2026

Place, Date

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